

RAI-Audit UK v1 - Loan Approval Governance Audit Report

Generated: 2026-05-11 06:23:05

Dataset: loan_approval_dataset_B.csv

Target: loan_status (0=Approved/Good, 1=Rejected/Bad)

Protected / proxy feature: age_group_audit (proxy=True)

Feature set: person_age, person_income, person_emp_length, loan_amnt, loan_int_rate, loan_percent_income, cb_person_cred_hist_length, flag_age_outlier, flag_emp_length_outlier, flag_person_emp_length_missing, flag_loan_int_rate_missing, log_person_income, log_loan_amnt, loan_to_income_recomputed, interest_burden_proxy, person_home_ownership_MORTGAGE, person_home_ownership_OTHER, person_home_ownership_OWN, person_home_ownership_RENT, loan_intent_DEBTCONSOLIDATION, loan_intent_EDUCATION, loan_intent_HOMEIMPROVEMENT, loan_intent_MEDICAL, loan_intent_PERSONAL, loan_intent_VENTURE, loan_grade_A, loan_grade_B, loan_grade_C, loan_grade_D, loan_grade_E, loan_grade_F, loan_grade_G, cb_person_default_on_file_N, cb_person_default_on_file_Y

Splits: train=22687, validation=4862, test=4862

Fairness audit note: The favourable outcome is class 0, meaning Approved/Good. For Dataset B age-group proxy auditing, AGE_UNDER_30 (unprivileged) is treated as the unprivileged audit group and AGE_30_PLUS (privileged) as the privileged audit group. SPD is calculated as the approval rate of the unprivileged group minus the approval rate of the privileged group. DI is calculated as the approval rate of the unprivileged group divided by the approval rate of the privileged group. EOD is calculated as the favourable-class true positive rate of the unprivileged group minus the favourable-class true positive rate of the privileged group. These metrics are fairness audit evidence only and do not prove legal discrimination or certify legal compliance.

GRSE interpretation note: The Governance Ready Scoring Engine is a risk score. A lower GRSE score indicates stronger governance readiness under the tested audit conditions.

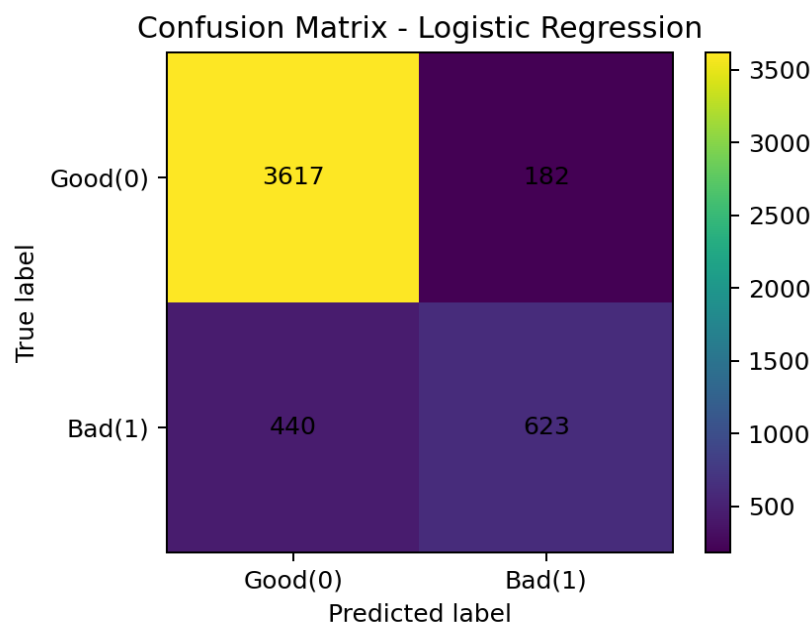
Model Card and Audit - Logistic Regression

Performance Score: 75.60
Governance Ready Scoring Engine (GRSE) Score: 21.75 (lower = stronger governance readiness)
GAR Classification: Green
Deployment Decision: Low Risk - Deployable

Predictive Performance

Metric	Validation	Test
Accuracy	0.869	0.872
Precision	0.763	0.774
Recall	0.582	0.586
F1	0.660	0.667
ROC-AUC	0.880	0.884

Confusion Matrix



Governance Evidence

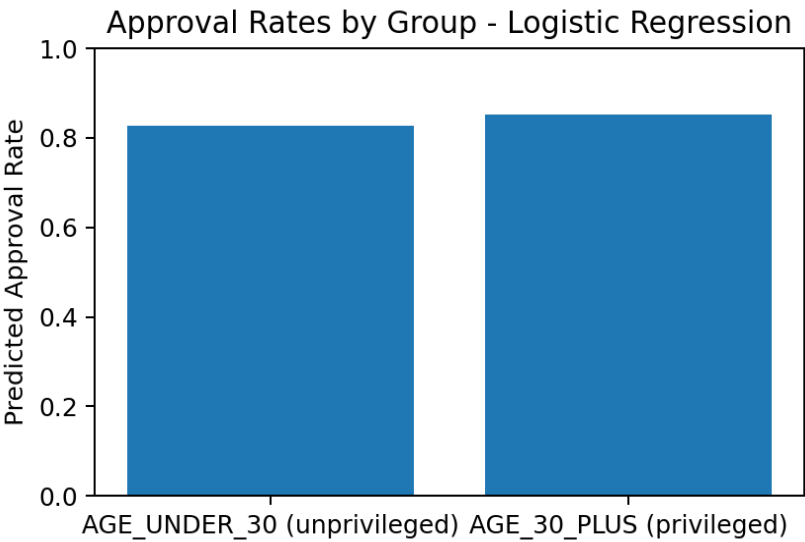
Metric	Value
Approval rate (AGE_UNDER_30 (unprivileged))	0.827
Approval rate (AGE_30_PLUS (privileged))	0.853
SPD: unprivileged - privileged	-0.026
DI: unprivileged / privileged	0.970
EOD: unprivileged - privileged	-0.004
Fairness risk	0.130
Flip-rate (noise)	0.007
Flip-rate (missingness)	0.005
Robustness risk	0.060

Metric	Value
Explainability risk	0.200
Top feature dominance	0.257

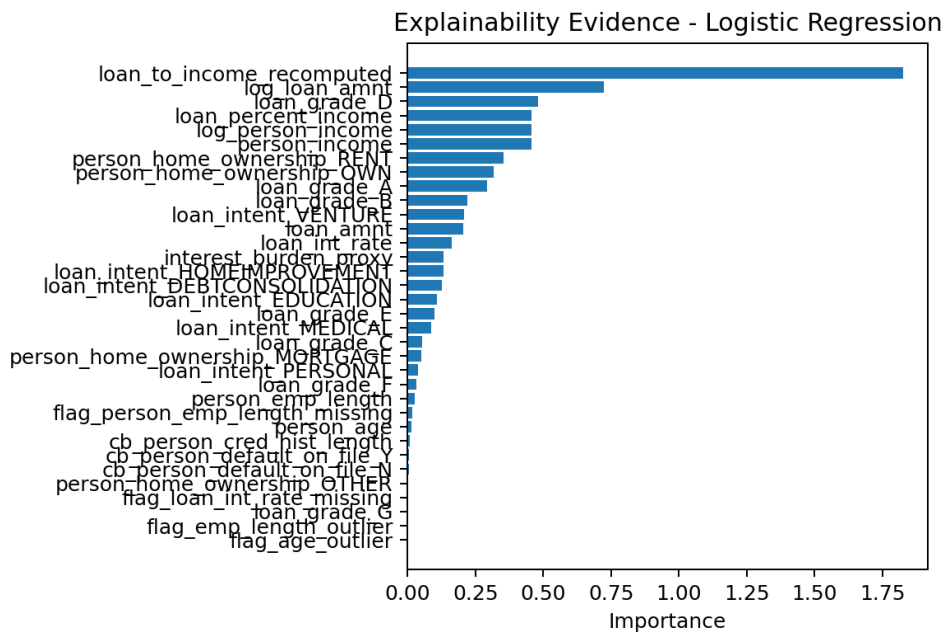
Governance Ready Scoring Engine (GRSE: lower score = stronger governance readiness)

GRSE Component	Risk Score Contribution
Fairness component	4.55
Robustness component	1.19
Explainability component	5.00
Governance baseline component	11.00
Total GRSE score (lower = stronger governance readiness)	21.75

Fairness Evidence



Explainability Evidence



Top Features

Feature	Importance
loan_to_income_recomputed	1.826566
log_loan_amnt	0.723303
loan_grade_D	0.480128
loan_percent_income	0.458162
log_person_income	0.457277
person_income	0.456563

Governance Mitigation Factors and Recommended Actions

Risk Area: Overall Governance

Identified factors:

- No material governance threshold was exceeded in this run

Recommended actions:

- Proceed only with controlled monitoring
- Repeat the audit on schedule and after material model changes

Evidence -> Regulation Traceability Matrix

Framework / Requirement	Evidence	Status
EU AI Act Art. 9 Risk Management	Risk score and mitigations generated	PASS
EU AI Act Art. 10 Data Governance / Bias	SPD, DI, EOD fairness evidence	PASS
EU AI Act Art. 13 Transparency	Explainability evidence generated	PASS
EU AI Act Art. 14 Human Oversight	Human-review mitigation included	PARTIAL
EU AI Act Art. 15 Robustness	Noise and missingness stress tests	PASS
UK FCA Principle 6	Group-level outcome monitoring and mitigations	PASS
UK FCA Principle 7	Reason-code style explanation artefacts	PASS
UK PRA Model Risk	Stress testing and governance documentation	PASS

Framework / Requirement	Evidence	Status
NIST AI RMF	Measure and Manage evidence included	PASS
ISO/IEC 42001	Governance and monitoring control suggestions	PARTIAL

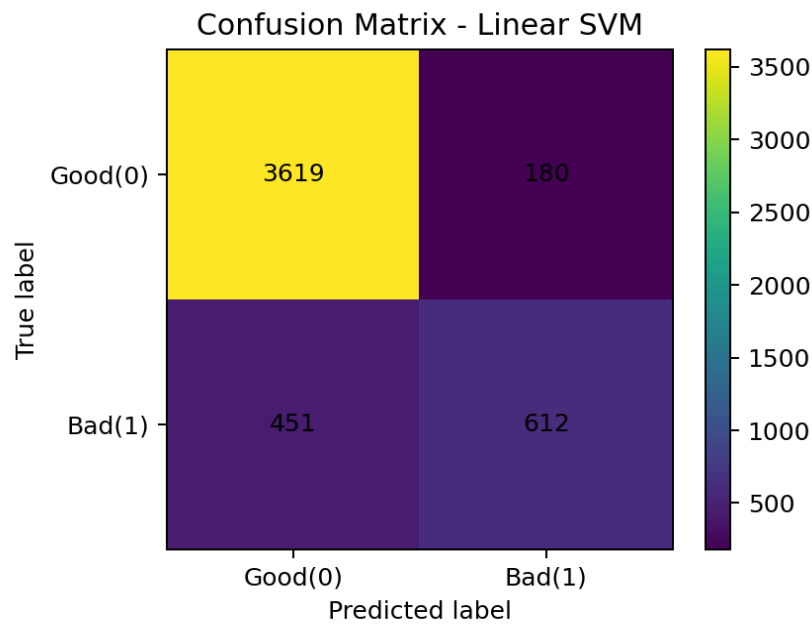
Model Card and Audit - Linear SVM

Performance Score: 75.16
Governance Ready Scoring Engine (GRSE) Score: 23.23 (lower = stronger governance readiness)
GAR Classification: Green
Deployment Decision: Low Risk - Deployable

Predictive Performance

Metric	Validation	Test
Accuracy	0.870	0.870
Precision	0.774	0.773
Recall	0.571	0.576
F1	0.657	0.660
ROC-AUC	0.879	0.882

Confusion Matrix



Governance Evidence

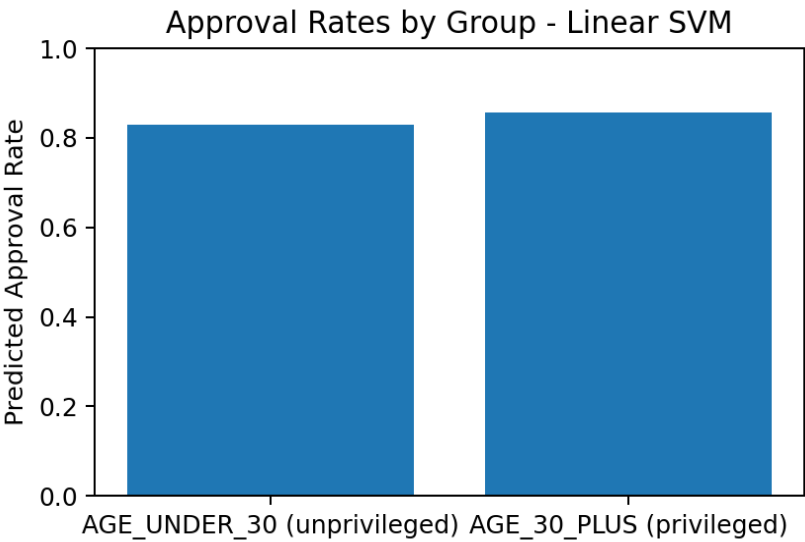
Metric	Value
Approval rate (AGE_UNDER_30 (unprivileged))	0.829
Approval rate (AGE_30_PLUS (privileged))	0.858
SPD: unprivileged - privileged	-0.028
DI: unprivileged / privileged	0.967
EOD: unprivileged - privileged	-0.004
Fairness risk	0.142
Flip-rate (noise)	0.007
Flip-rate (missingness)	0.003
Robustness risk	0.050

Metric	Value
Explainability risk	0.250
Top feature dominance	0.222

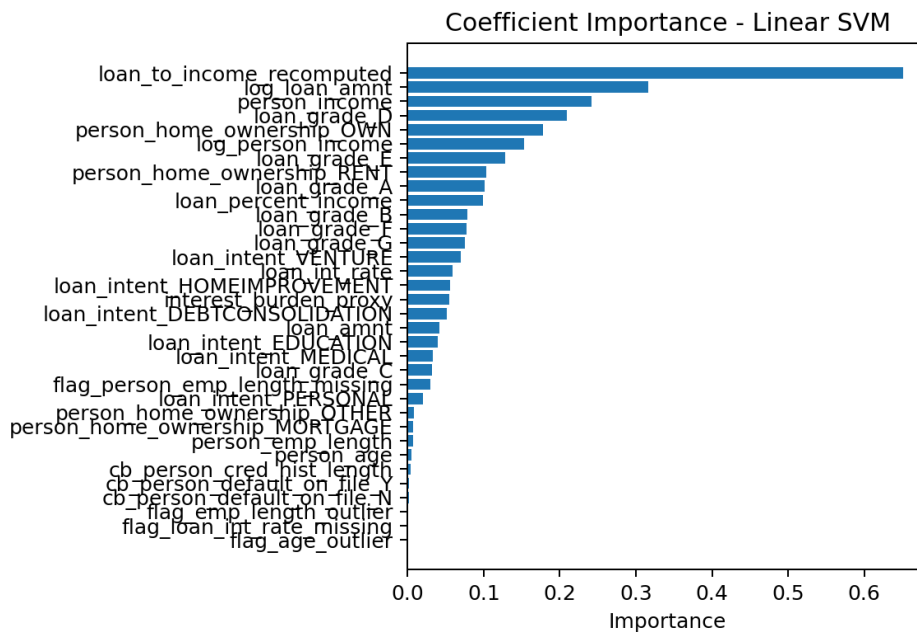
Governance Ready Scoring Engine (GRSE: lower score = stronger governance readiness)

GRSE Component	Risk Score Contribution
Fairness component	4.97
Robustness component	1.01
Explainability component	6.25
Governance baseline component	11.00
Total GRSE score (lower = stronger governance readiness)	23.23

Fairness Evidence



Explainability Evidence



Top Features

Feature	Importance
loan_to_income_recomputed	0.650829
log_loan_amnt	0.316005
person_income	0.241825
loan_grade_D	0.208871
person_home_ownership_OWN	0.178323
log_person_income	0.152932

Governance Mitigation Factors and Recommended Actions

Risk Area: Overall Governance

Identified factors:

- No material governance threshold was exceeded in this run

Recommended actions:

- Proceed only with controlled monitoring
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Evidence -> Regulation Traceability Matrix

Framework / Requirement	Evidence	Status
EU AI Act Art. 9 Risk Management	Risk score and mitigations generated	PASS
EU AI Act Art. 10 Data Governance / Bias	SPD, DI, EOD fairness evidence	PASS
EU AI Act Art. 13 Transparency	Explainability evidence generated	PASS
EU AI Act Art. 14 Human Oversight	Human-review mitigation included	PARTIAL
EU AI Act Art. 15 Robustness	Noise and missingness stress tests	PASS
UK FCA Principle 6	Group-level outcome monitoring and mitigations	PASS
UK FCA Principle 7	Reason-code style explanation artefacts	PASS
UK PRA Model Risk	Stress testing and governance documentation	PASS

Framework / Requirement	Evidence	Status
NIST AI RMF	Measure and Manage evidence included	PASS
ISO/IEC 42001	Governance and monitoring control suggestions	PARTIAL

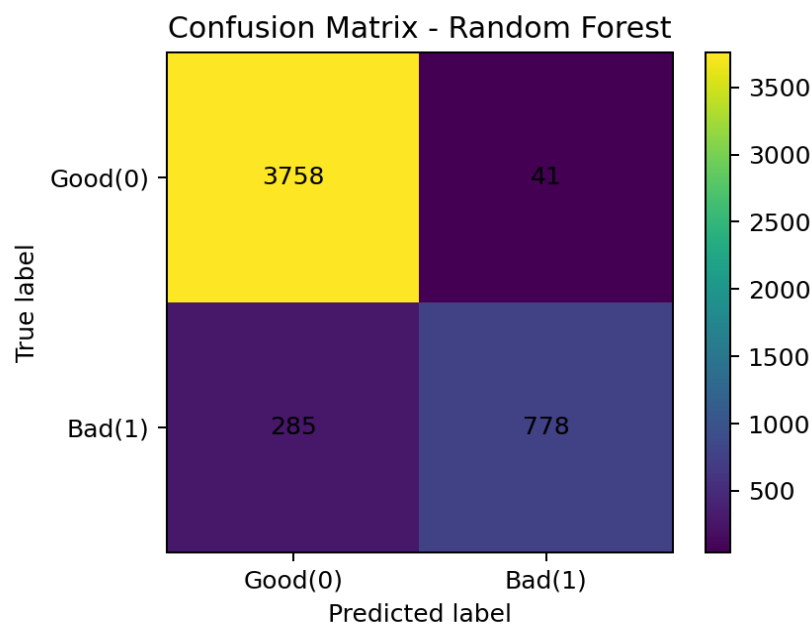
Model Card and Audit - Random Forest

Performance Score: 87.57
Governance Ready Scoring Engine (GRSE) Score: 23.71 (lower = stronger governance readiness)
GAR Classification: Green
Deployment Decision: Low Risk - Deployable

Predictive Performance

Metric	Validation	Test
Accuracy	0.933	0.933
Precision	0.957	0.950
Recall	0.727	0.732
F1	0.826	0.827
ROC-AUC	0.936	0.938

Confusion Matrix



Governance Evidence

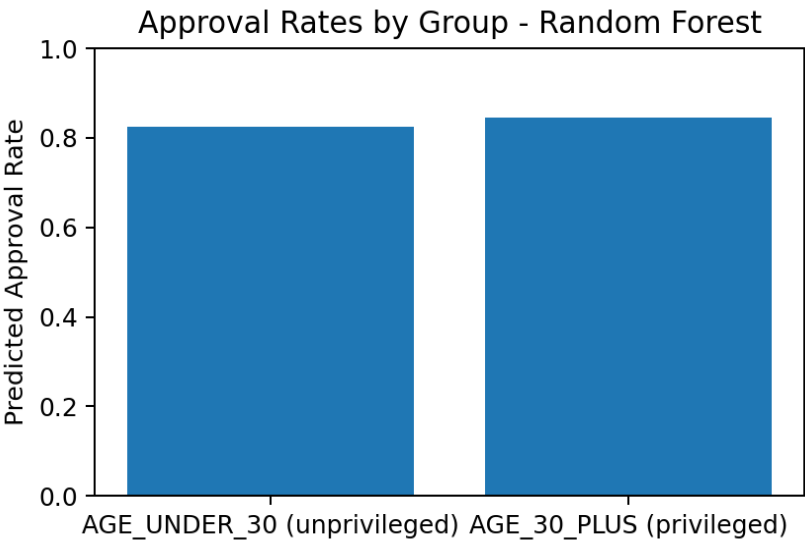
Metric	Value
Approval rate (AGE_UNDER_30 (unprivileged))	0.826
Approval rate (AGE_30_PLUS (privileged))	0.846
SPD: unprivileged - privileged	-0.020
DI: unprivileged / privileged	0.977
EOD: unprivileged - privileged	-0.006
Fairness risk	0.099
Flip-rate (noise)	0.003
Flip-rate (missingness)	0.001
Robustness risk	0.025

Metric	Value
Explainability risk	0.350
Top feature dominance	0.116

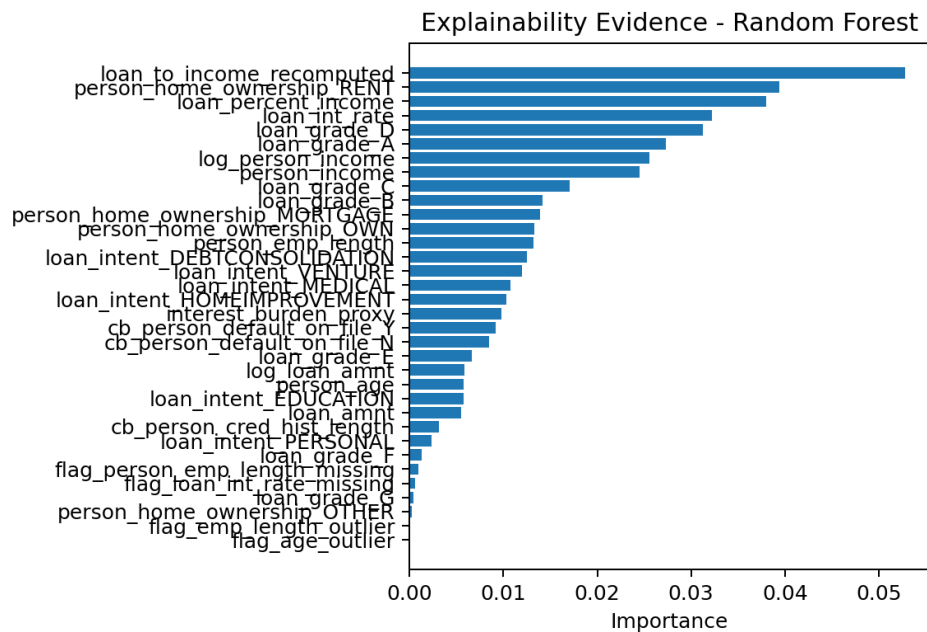
Governance Ready Scoring Engine (GRSE: lower score = stronger governance readiness)

GRSE Component	Risk Score Contribution
Fairness component	3.47
Robustness component	0.49
Explainability component	8.75
Governance baseline component	11.00
Total GRSE score (lower = stronger governance readiness)	23.71

Fairness Evidence



Explainability Evidence



Top Features

Feature	Importance
loan_to_income_recomputed	0.052795
person_home_ownership_RENT	0.039422
loan_percent_income	0.038007
loan_int_rate	0.032244
loan_grade_D	0.031233
loan_grade_A	0.027310

Governance Mitigation Factors and Recommended Actions

Risk Area: Overall Governance

Identified factors:

- No material governance threshold was exceeded in this run

Recommended actions:

- Proceed only with controlled monitoring
- Repeat the audit on schedule and after material model changes

Evidence -> Regulation Traceability Matrix

Framework / Requirement	Evidence	Status
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EU AI Act Art. 13 Transparency	Explainability evidence generated	PASS
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EU AI Act Art. 15 Robustness	Noise and missingness stress tests	PASS
UK FCA Principle 6	Group-level outcome monitoring and mitigations	PASS
UK FCA Principle 7	Reason-code style explanation artefacts	PASS
UK PRA Model Risk	Stress testing and governance documentation	PASS

Framework / Requirement	Evidence	Status
NIST AI RMF	Measure and Manage evidence included	PASS
ISO/IEC 42001	Governance and monitoring control suggestions	PARTIAL

Comparative Evaluation - Performance-Centric vs Governance-Centric Ranking

Ranking note: Performance score is ranked highest to lowest. GRSE is a risk score, so governance score is ranked lowest to highest.

Model	Performance Score	Performance Rank	GRSE Score (lower = stronger)	Governance Rank	Ranking Shift
Random Forest	87.57	1	23.71	3	-2
Logistic Regression	75.60	2	21.75	1	1
Linear SVM	75.16	3	23.23	2	1

Mitigation Report - Why Loans Were Not Approved (Dataset-Driven)

Logistic Regression: Common Rejection Drivers

Feature	Effect (Rejected vs Approved)	Direction	Human-readable reason
loan_to_income_recomputed	1.305	higher	loan_to_income_recomputed
loan_percent_income	1.300	higher	loan_percent_income
loan_grade_D	1.208	higher	loan_grade_D
loan_int_rate	1.066	higher	loan_int_rate
log_person_income	-0.924	lower	log_person_income
interest_burden_proxy	0.703	higher	interest_burden_proxy

Example applicant-facing reasons (sample rejected cases)

- Case #3: loan_grade_F; loan_int_rate; log_loan_amnt
- Case #5: loan_grade_D; loan_int_rate; person_home_ownership_RENT
- Case #11: loan_to_income_recomputed; loan_grade_D; loan_int_rate
- Case #13: loan_grade_D; log_loan_amnt; log_person_income
- Case #23: loan_to_income_recomputed; loan_grade_D; loan_amnt
- Case #24: loan_grade_D; loan_intent_HOMEIMPROVEMENT; loan_to_income_recomputed
- Case #27: loan_to_income_recomputed; log_person_income; person_home_ownership_RENT
- Case #30: log_person_income; loan_to_income_recomputed; log_loan_amnt

Suggested applicant mitigation steps

- General guidance: strengthen credit profile, reduce exposure, and improve supporting evidence.

Linear SVM: Common Rejection Drivers

Feature	Effect (Rejected vs Approved)	Direction	Human-readable reason
loan_to_income_recomputed	1.298	higher	loan_to_income_recomputed
loan_percent_income	1.293	higher	loan_percent_income
loan_grade_D	1.273	higher	loan_grade_D
loan_int_rate	1.105	higher	loan_int_rate
log_person_income	-0.871	lower	log_person_income
interest_burden_proxy	0.730	higher	interest_burden_proxy

Example applicant-facing reasons (sample rejected cases)

- Case #3: loan_grade_F; loan_int_rate; log_loan_amnt
- Case #5: loan_grade_D; loan_int_rate; person_home_ownership_RENT
- Case #11: loan_to_income_recomputed; loan_grade_D; loan_int_rate
- Case #13: loan_grade_D; log_loan_amnt; log_person_income
- Case #23: loan_to_income_recomputed; loan_grade_D; loan_grade_A
- Case #24: loan_grade_D; loan_intent_HOMEIMPROVEMENT; loan_to_income_recomputed
- Case #27: loan_to_income_recomputed; log_person_income; person_home_ownership_RENT

- Case #30: loan_to_income_recomputed; log_person_income; log_loan_amnt

Suggested applicant mitigation steps

- General guidance: strengthen credit profile, reduce exposure, and improve supporting evidence.

Random Forest: Common Rejection Drivers

Feature	Effect (Rejected vs Approved)	Direction	Human-readable reason
loan_percent_income	1.172	higher	loan_percent_income
loan_to_income_recomputed	1.168	higher	loan_to_income_recomputed
loan_grade_D	1.072	higher	loan_grade_D
loan_int_rate	0.933	higher	loan_int_rate
log_person_income	-0.892	lower	log_person_income
person_home_ownership_RENT	0.730	higher	person_home_ownership_RENT

Example applicant-facing reasons (sample rejected cases)

- Case #3: loan_percent_income (higher; value=0); loan_to_income_recomputed (higher; value=0); loan_grade_D (higher; value=0)

- Case #5: loan_percent_income (higher; value=0); loan_to_income_recomputed (higher; value=0); loan_grade_D (higher; value=1)

- Case #13: loan_percent_income (higher; value=0); loan_to_income_recomputed (higher; value=0); loan_grade_D (higher; value=1)

- Case #14: loan_percent_income (higher; value=0); loan_to_income_recomputed (higher; value=0); loan_grade_D (higher; value=0)

- Case #27: loan_percent_income (higher; value=0); loan_to_income_recomputed (higher; value=0); loan_grade_D (higher; value=0)

- Case #30: loan_percent_income (higher; value=0); loan_to_income_recomputed (higher; value=0); loan_grade_D (higher; value=0)

- Case #33: loan_percent_income (higher; value=0); loan_to_income_recomputed (higher; value=0); loan_grade_D (higher; value=0)

- Case #34: loan_percent_income (higher; value=0); loan_to_income_recomputed (higher; value=0); loan_grade_D (higher; value=1)

Suggested applicant mitigation steps

- General guidance: strengthen credit profile, reduce exposure, and improve supporting evidence.